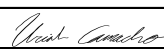


☐ High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. Project No.: WW-4705R
Project Name: Hilo WWTP Phase 1 Submittal Number: 16133.002.1
Submittal Title: Duct Banks ☐ For Information Only
TO: County
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec: 16133	Paragraph: N/a
Authorized By: CMC Rebar	Date Submitted: 09/19/2025

Submittal Certification	
Check Either (A) or (B):	
☒ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> .
☐ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.	
General Contractor's Reviewer's Signature: 	
Printed Name and Title: Uriah Camacho, Project Engineer	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm:	Signature: Date Returned:

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 9/19/2025
SPECIFICATION SECTION # 16133
SPECIFICATION Duct Banks
PARAGRAPH n/a
DRAWING CMC
SUBCONTRACTOR n/a
SUPPLIER n/a
MANUFACTURER n/a

CERTIFIED BY CQCM or Designee : 



Submittal Review Response

Project Name: Hilo WWTP Rehabilitation and Replacement Project Phase 1
Submittal No.: 16133-002.0
Date: [Publish Date]

Client: County of Hawai'i Carollo Project No.: 203975
Contractor: Nan, Inc.
Submittal Name: Duct Banks shop drawing
Reviewed By: Kellen Gross, Marissa Kurniawan, Hipom Caleb Che, Bruce DiFrancisco

SUBMITTAL REVIEW

Review is for general compliance with contract documents. No responsibility is assumed by Carollo for correctness of quantities, dimensions, and details. No deviation or variation is approved unless specifically addressed in these review comments. Refer to Section 01330 for additional requirements. The Contractor shall assume full responsibility for coordination with all other trades and deviations from contract requirements.

Approved	☐	No Exceptions
	☐	Make Corrections Noted - See Comments
	☐	Make Corrections Noted - Confirm
Not Approved	☒	Correct and Resubmit
	☐	Rejected - See Remarks
Receipt Acknowledged	☐	Filed for Record
	☐	With Comments - Resubmit

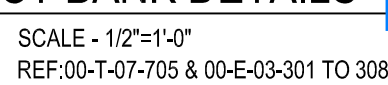
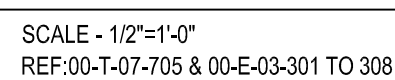
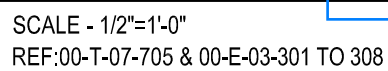
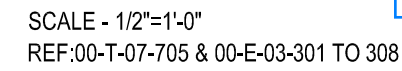
Review Comments:

1. Comments are provided in the attachment. Verify and address all comments.
2. Missing the following duct banks. Provide these dimensions in resubmittal. Addressed
 - a. 10B
 - b. 11G
 - c. 11H
 - d. 21I
1. 4-conduit duct bank serving solids odor control (2'-4"W x 1'-5"H). See drawing 14-E-01-101. Addressed
3. Identify duct bank sections on drawings RP-012 and RP-013. Addressed
4. Minimum lap splicing of 1'-4" required for U-bar stirrups per detail EM001/TYP, Keynote 7. Verify and revise the submittal as necessary to provide the required lap splicing, while also keeping the height of the ductbanks and the minimum 3" concrete cover at top and bottom. Addressed
2. Submitted PDF appears to be missing rebar details at corner bends. Provide details at corner bends following configurations shown on detail S144/TYP. Addressed
3. Drawing RP-011 is accepted as a schematic representation of the duct bank runs that will be installed on the site, with

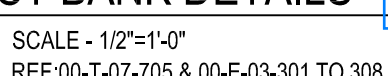
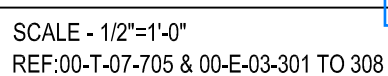
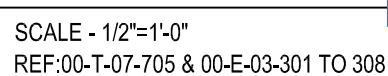
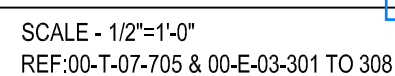
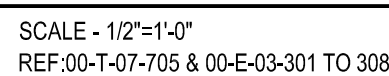
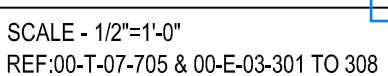
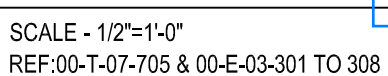
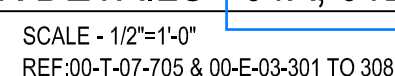
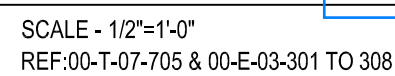
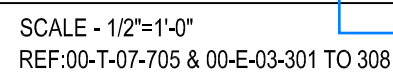
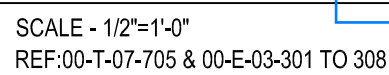
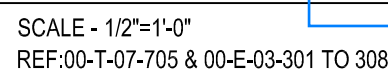
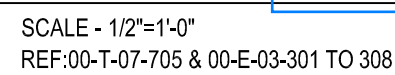
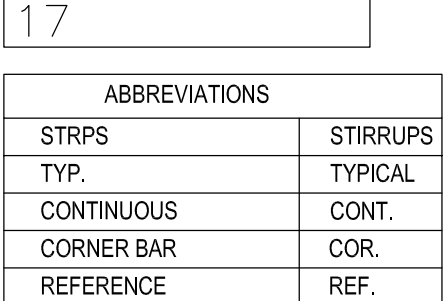
the caveat that comments provided in the attachment are addressed. No Comment

4. It is understood that a submittal that locates the duct banks in space, either via dimensioning against fixed structures or by coordinate location, will be provided under separate cover. No Comment

5. It is understood that duct bank profiles with the location of all underground utilities will be provided under separate cover in accordance with Section 1.02.C.1. No Comment



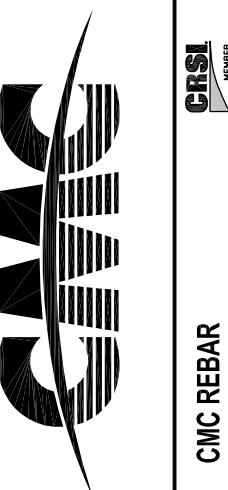
Bar list produced by RebarCAD 2020.0



ALL ITEMS THAT ARE CLOUDED &/OR MARKED "VERY" REFER TO QUESTIONS, OMISSIONS, CONFLICTS OR ERRORS IN THE CONTRACT DOCUMENTS. UNLESS CORRECTIVE ACTION IS INDICATED ON THIS APPROVAL, ALL ITEMS CLOUDED WILL BE ASSUMED CORRECT AS SUBMITTED. NO FURTHER CORRECTIONS WILL BE ALLOWED. CMC REBAR ASSUMES NO RESPONSIBILITY FOR ERRORS CAUSED BY LACK OF REQUESTED INFORMATION.

DATE: 06/20/25	FOR: APPROVAL
DATE: .	FOR: .
DATE: .	FOR: .
DATE: 08/13/25	REVISD AS PER MAIL COMMENTS
DATE: 09/19/25	REVISD AS PER A/E COMMENTS
DATE: .	DATE: .

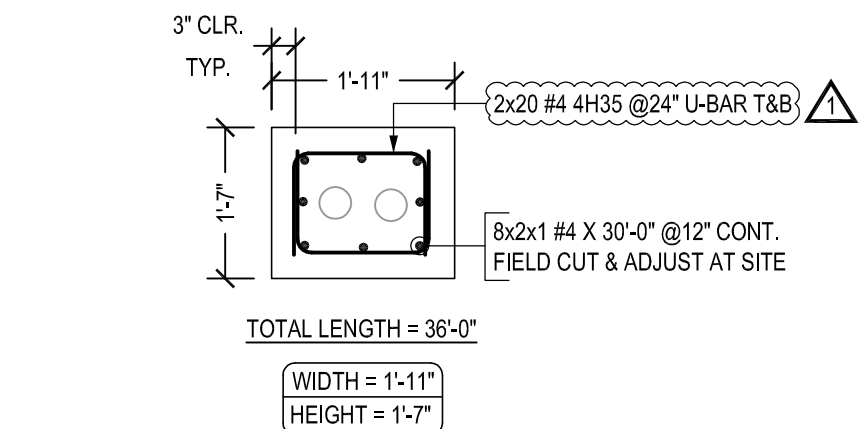
CUSTOMER:	-	ORDER #	257424001
ORDER NAME	HLO WWP	ESTIMATOR	MGR
		EMAIL:	will@mcmasterbar.com
LOCATION:	COUNTY OF HAWAII	PHONE:	(360)-346-5729
EXCEPTION:	ELECTRICAL DUCT BANK REINFORCEMENT DETAILS		
		REF:	AS NOTED



2524240001

WING:

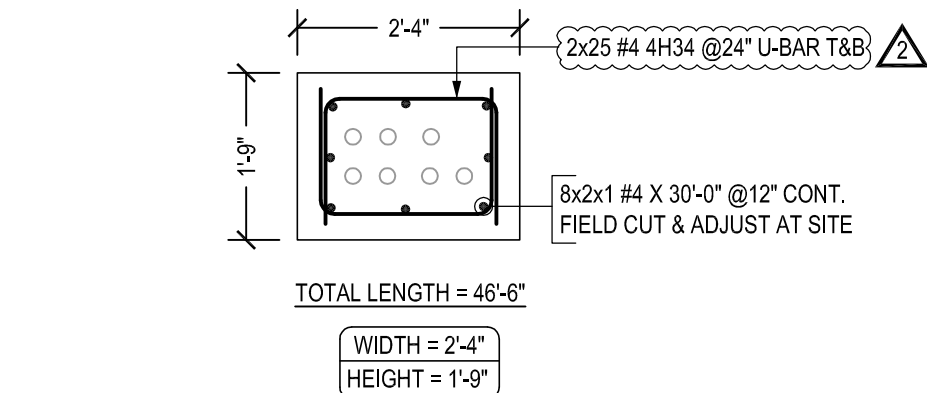
RP-012



TYPICAL DUCT BANK DETAILS - 21L

SCALE - 1/2"=1'-0"

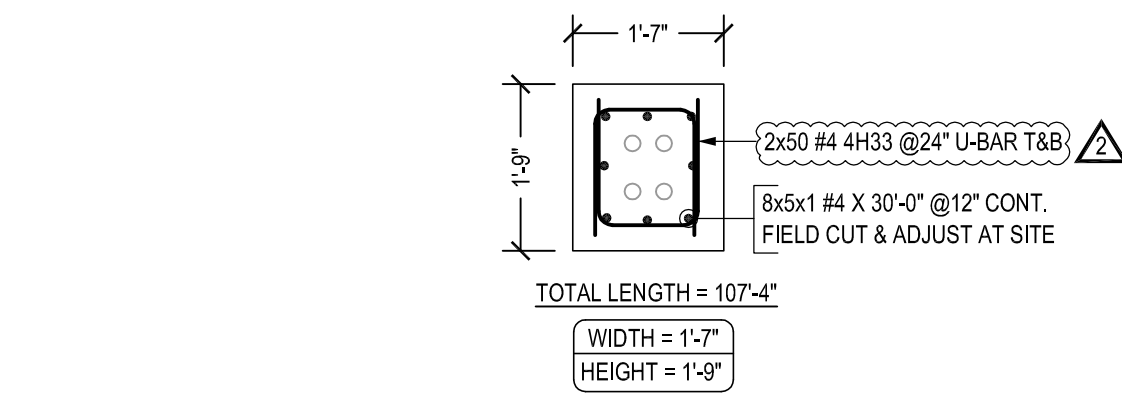
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TYPICAL DUCT BANK DETAILS - 11C & 14B

SCALE - 1/2"=1'-0"

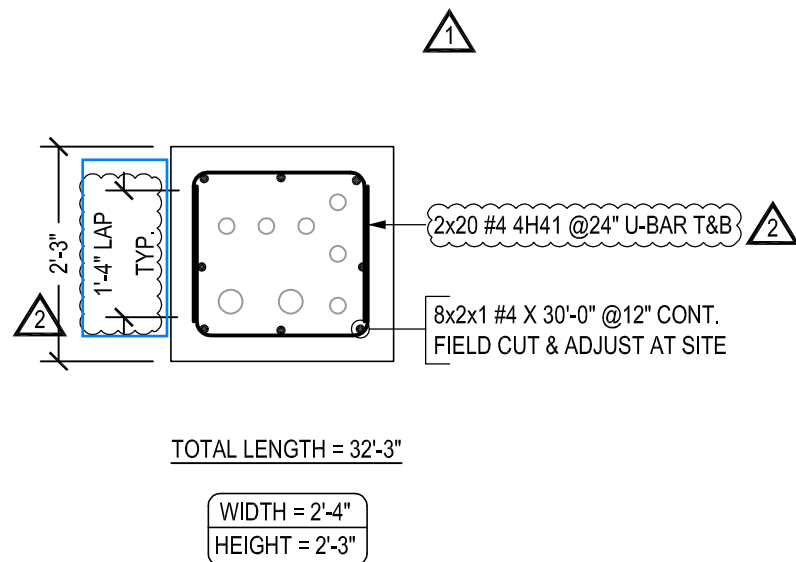
REF:00-T-07-705 & 00-E-03-301 TO 308



TYPICAL DUCT BANK DETAILS - 23A, 21F, 13A, 11D, 11G, 11H & 11I

SCALE - 1/2"=1'-0"

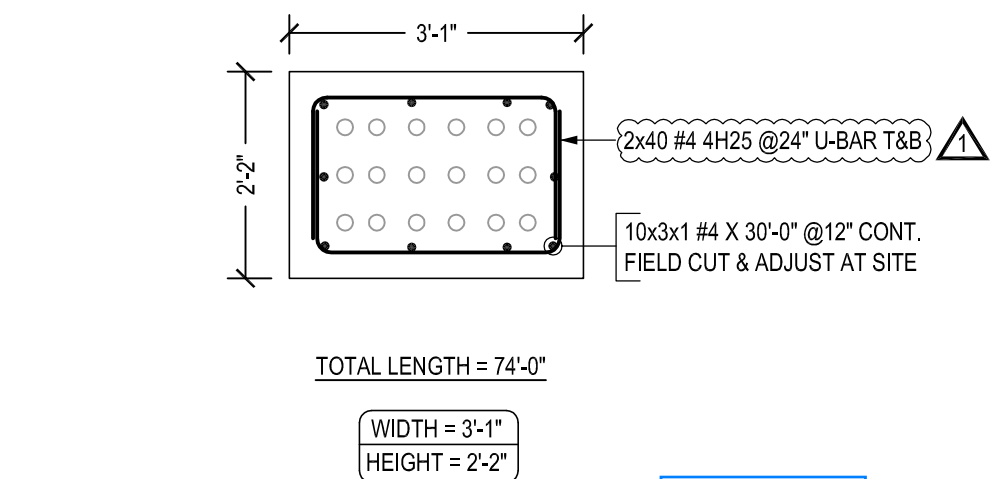
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TYPICAL DUCT BANK DETAILS - 13B

SCALE - 1/2"=1'-0"

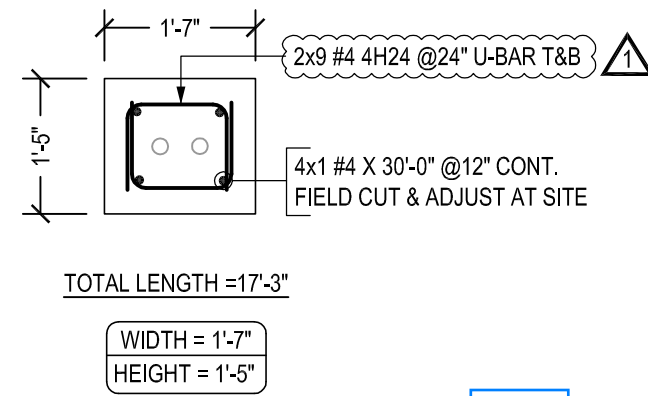
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TYPICAL DUCT BANK DETAILS - 14A & 21A

SCALE - 1/2"=1'-0"

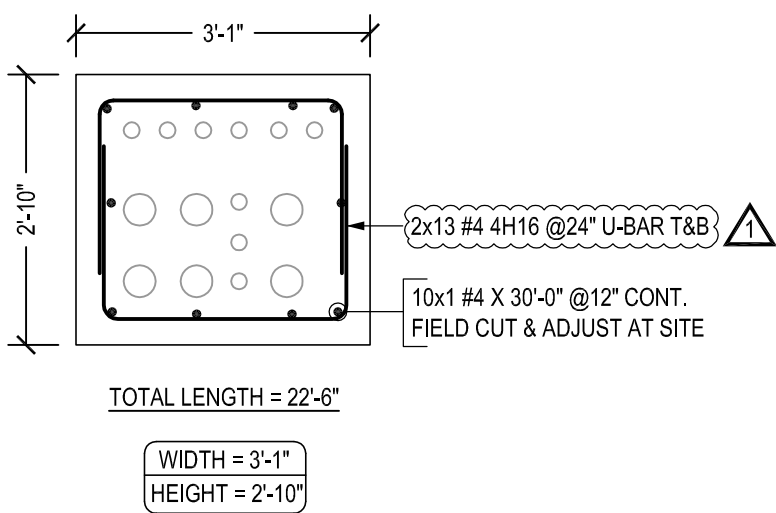
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TYPICAL DUCT BANK DETAILS - 13C

SCALE - 1/2"=1'-0"

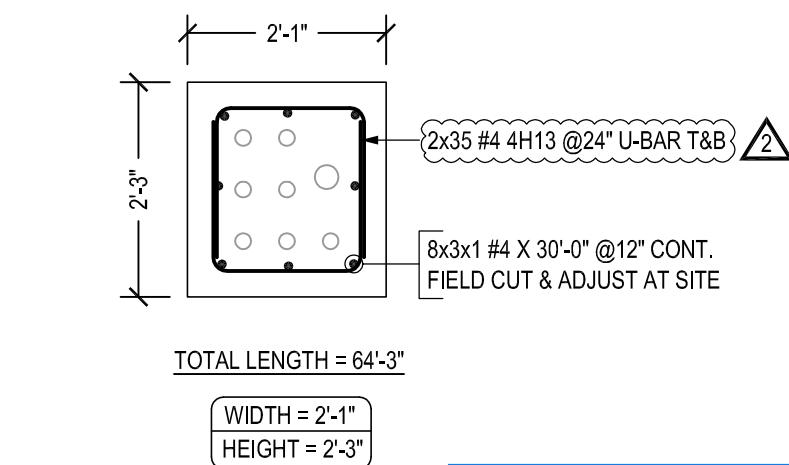
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TYPICAL DUCT BANK DETAILS - 25A

SCALE - 1/2"=1'-0"

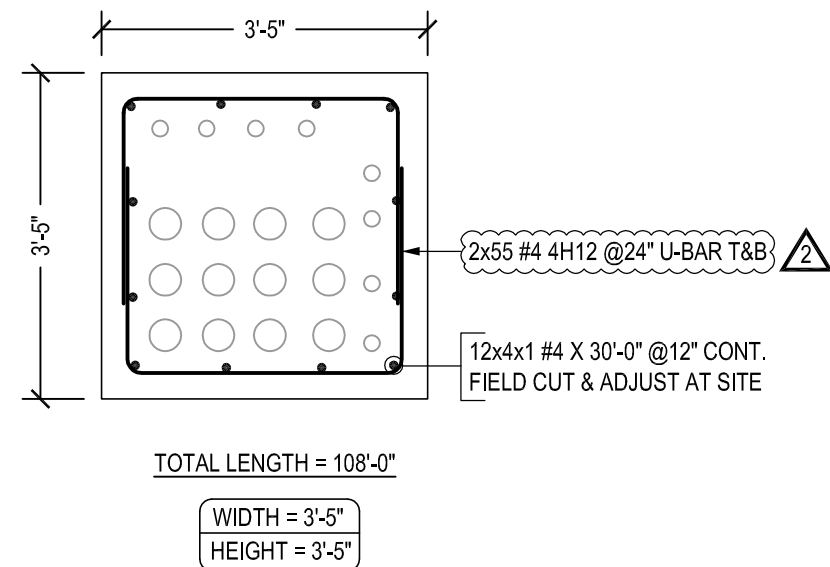
REF:00-T-07-705 & 00-E-03-301 TO 308



TYPICAL DUCT BANK DETAILS - 06C, 06E & 06G

SCALE - 1/2"=1'-0"

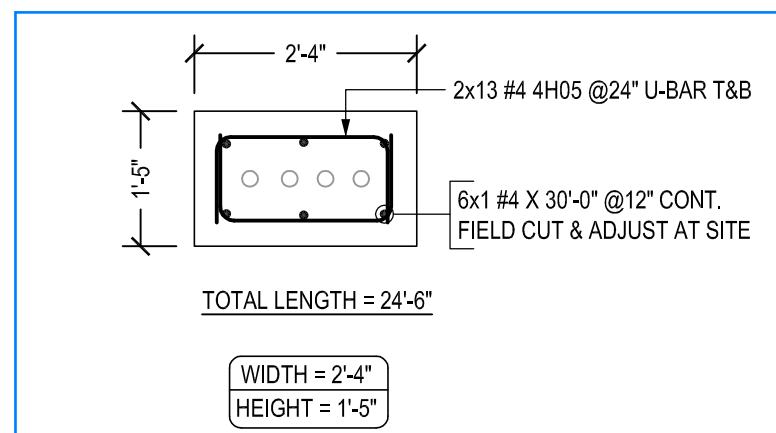
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TYPICAL DUCT BANK DETAILS - 12A

SCALE - 1/2"=1'-0"

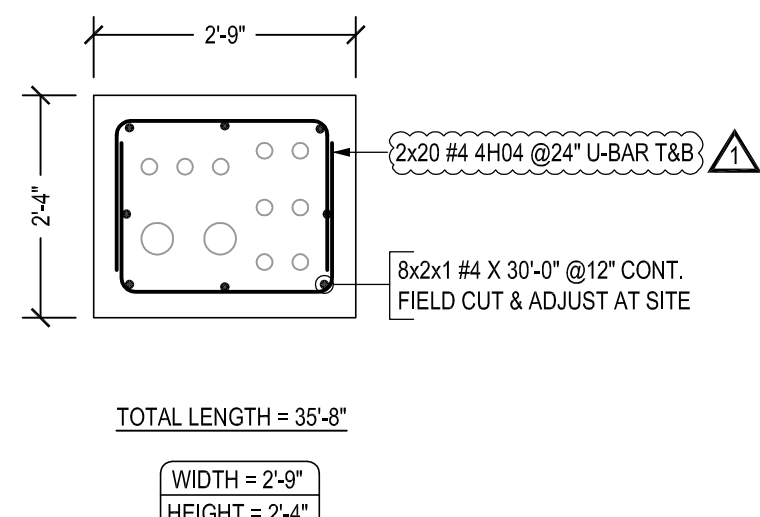
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TYPICAL DUCT BANK DETAILS - 05B

SCALE - 1/2"=1'-0"

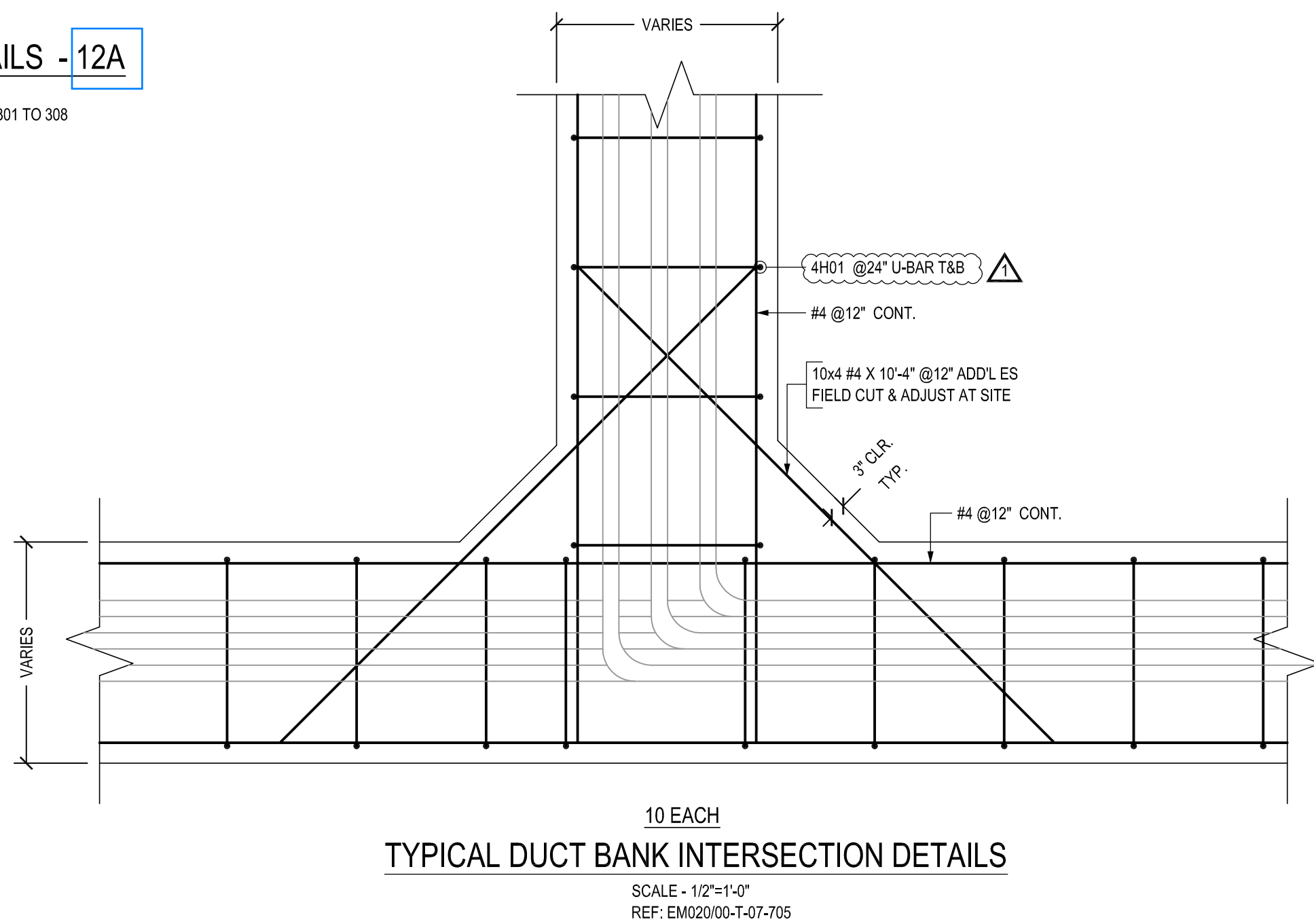
REF:00-T-07-705 & 00-E-03-301 TO 308



TYPICAL DUCT BANK DETAILS - 09A

SCALE - 1/2"=1'-0"

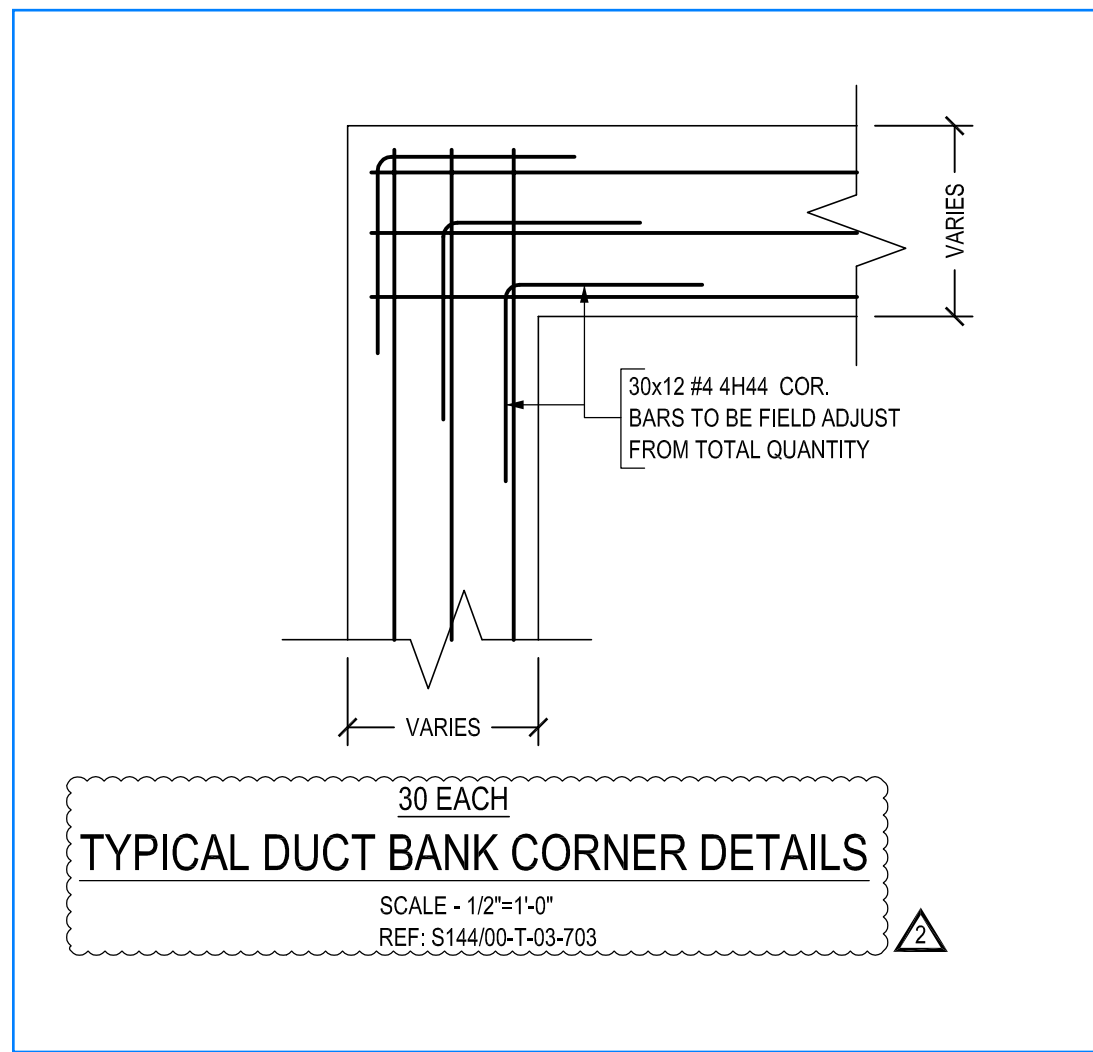
REF:00-T-07-705 & 00-E-03-301 TO 308



TYPICAL DUCT BANK INTERSECTION DETAILS

SCALE - 1/2"=1'-0"

REF: EM020/00-T-07-705



TYPICAL DUCT BANK CORNER DETAILS

SCALE - 1/2"=1'-0"

REF: S144/00-T-03-703

Release Number: RP-013				BAR LIST															
Bar Mark	Qty	Size	Total Length	Type	'A'	'B'	'C'	'D'	'E'	'F'	'G'	'H'	'J'	'K'	'O'	'R'			
4H04	40	#4	5'-5"	17		1'-7"	2'-3"	1'-7"											
4H05	26	#4	3'-8"	17		0'-11"	1'-10"	0'-11"											
4H12	110	#4	7'-3"	17		2'-2"	2'-11"	2'-2"											
4H13	70	#4	4'-9"	17		1'-7"	1'-7"	1'-7"											
4H16	26	#4	6'-3"	17		1'-10"	2'-7"	1'-10"											
4H24	18	#4	2'-11"	17		0'-11"	1'-1"	0'-11"											
4H25	80	#4	5'-7"	17		1'-6"	2'-7"	1'-6"											
4H33	100	#4	3'-9"	17		1'-4"	1'-1"	1'-4"											
4H34	50	#4	4'-6"	17		1'-4"	1'-10"	1'-4"											
4H35	40	#4	3'-7"	17		1'-1"	1'-5"	1'-1"											
4H41	40	#4	5'-0"	17		1'-7"	1'-10"	1'-7"											
4H44	360	#4	4'-2"	17		2'-1"	2'-1"												

Bar list produced by RebarCAD 2020.0

Bar list produced by RebarCAD 2020.0

ABBREVIATIONS	
BOTTOM	BOT.
TOP&BOTTOM	T&B
CONTINUOUS	CONT.
CORNER BAR	COR.
REFERENCE	REF.

LAP SCHEDULE		
SIZE	4000 PSI	
	TOP	OTHERS
#4	32"	25"

FIELD USE:
THIS DRAWING IS TO BE USED IN CONJUNCTION WITH THE STRUCTURAL AND ARCHITECTURAL DOCUMENTS TO FACILITATE PLACING AND TIEING OF REINFORCEMENT. ANY DIMENSIONS SHOWN ON THIS DRAWING ARE TO BE USED FOR CONSTRUCTION. ANY DIMENSIONS OF REINFORCEMENT NOT SHOWN ON THIS DRAWING ARE TO BE USED FOR CONSTRUCTION. ANY DIMENSIONS OF REINFORCEMENT NOT SHOWN ON THIS DRAWING ARE TO BE USED FOR CONSTRUCTION. ANY DIMENSIONS OF REINFORCEMENT NOT SHOWN ON THIS DRAWING ARE TO BE USED FOR CONSTRUCTION.

APPROVAL AUTHORITY:
THIS DRAWING IS TO BE USED IN CONJUNCTION WITH THE STRUCTURAL AND ARCHITECTURAL DOCUMENTS TO FACILITATE PLACING AND TIEING OF REINFORCEMENT. ANY DIMENSIONS SHOWN ON THIS DRAWING ARE TO BE USED FOR CONSTRUCTION. ANY DIMENSIONS OF REINFORCEMENT NOT SHOWN ON THIS DRAWING ARE TO BE USED FOR CONSTRUCTION. ANY DIMENSIONS OF REINFORCEMENT NOT SHOWN ON THIS DRAWING ARE TO BE USED FOR CONSTRUCTION.

SUBMITTAL RECORD	
DATE	FOR
16/70/25	APPROVAL

PROJECT INFORMATION	
JOB #	252424001
REBAR	W/C
DATE	16/70/25

PROJECT INFORMATION	
JOB #	252424001
REBAR	W/C
DATE	16/70/25

PROJECT INFORMATION	
JOB #	252424001
REBAR	W/C
DATE	16/70/25

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JOB #	252424001
REBAR	W/C
DATE	16/70/25

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REBAR	W/C
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